

LEARNING SPARK

A Project Report

Submitted by

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in partial fulfilment for the award of the degree of

BACHELOR OF COMPUTER APPLICATIONS

Under the supervision of

Asst. Prof. Mr. JOHNBENEDICT G



DON BOSCO COLLEGE(CO-ED)

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DEPARTMENT OF COMPUTER APPLICATIONS

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CANDIDATE'S DECLARATION

I hereby certify that the project entitled “**LEARNING SPARK**” submitted by PRAVEEN Y(35822U09078) partial fulfillment of the requirement for the award of degree of the B. C.A submitted at DON BOSCO COLLEGE(CO-ED) YELAGIRI HILLS is an authentic record of my own work carried out during a period from 12/12/2024 to 25/03/2025 under the guidance of Asst. prof. Mr. JOHN BENADICT (M.C.A, M.Phil,SET), Department of Computer Application. The matter presented in this project has not formed the basis for the award of any other degree, diploma, fellowship or any other similar titles.

Signature of the Student

Place:

Date:

CERTIFICATE

This is to certify that the project titled “ **LEARNING SPARK**” is the bonafide work carried out by PRAVEEN Y (35822U09078) in partial fulfillment of the requirement for the award of degree of the B.C.A submitted at DON BOSCO COLLEGE (CO-ED) YELAGIRI HILLS is an authentic record his/her work carried out during a period from 12/12/1024 to 25/03/2025 under the guidance of Asst. prof. Mr. JOHN BENADICT, Department of Computer Applications . The Major Project Viva-Vice Examination has been held on _____.

Signature of the Guide

Signature of the HOD

Internal Examiner

External Examiner

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ABSTRACT

“LEARNING SPARK”! This is my web-based Learning Management System (LMS) that’s all about making education fun and interactive. It’s built around three main roles: Users (the learners), Staff (the content creators), and one awesome Admin named Praveen. Praveen’s got the reins—he can add, update, or remove users and staff, plus keep an eye on how everything’s going with cool stats. Staff get their own special pages to upload videos, documents, and daily blogs, and they can chat with users and Praveen in real time. Users, meanwhile, can check out content, download it, ask questions, pick their favorite staff, and even read blogs—all from their own unique pages. It’s designed to spark learning with real-time chats and personalized vibes, making it way better than old-school platforms.

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CHAPTER – I

INTRODUCTION

1.1 INTRODUCTION

Learning Spark is a web-based Learning Management System (LMS) that provides an interactive and engaging learning experience for students and PhD educators. Unlike traditional LMS platforms, Learning Spark introduces real-time communication and personalized user experiences to enhance knowledge sharing. The platform consists of three main roles, Admin: Manages users, monitors platform activities, and maintains the system. Staff (Educators): Uploads videos, documents, and blogs while interacting with students. Users (Students): Access educational materials, engage in discussions, and receive feedback from educators.

1.1.1 Problem Statements:

Traditional online learning platforms lack real-time interaction, making it difficult for students to engage effectively with educators. Many existing Learning Management Systems (LMS) are static, focusing only on content delivery without allowing active communication. Additionally, educators struggle to track student engagement and provide personalized feedback. The absence of interactive features results in a disconnected learning experience, leading to reduced motivation and effectiveness in knowledge transfer.

1.1.2 Problem Solutions:

Learning Spark addresses these challenges by integrating real-time messaging, personalized user pages, and data analytics for better tracking of student engagement. By implementing a structured system with three distinct roles (Admin, Staff, and User), the platform enhances interaction and content management. The system provides. Live chat functionality for immediate communication between students and educators. Personalized dashboards where users can access and manage their content. Advanced analytics to help educators monitor engagement and optimize content delivery. User-friendly UI/UX with responsive design to ensure accessibility across different devices. By combining these features, Learning Spark fosters a dynamic and interactive learning environment, making online education more engaging and effective.

1.2 PROJECT PROPOSAL

1.2.1 Title of the Project: LEARNING SPARK

1.2.2 Problem Description:

"LEARNING SPARK" is a response to significant inefficiencies plaguing the current learning management systems (LMS) at Don Bosco College (Co-Ed), Yelagiri Hills. These systems, whether traditional or digital, fail to meet the evolving needs of students, staff, and administrators in a technology-driven educational landscape. The primary challenge lies in the reliance on outdated methods—such as static content delivery through uploaded PDFs or pre-recorded lectures—which offer little to no real-time interaction. This lack of dynamic engagement leaves students disconnected from educators, unable to ask questions or seek instant clarifications, ultimately reducing their motivation and understanding of complex topics.

Additionally, the existing platforms provide a generic user experience, with interfaces that do not differentiate between the needs of administrators, staff, and students. This one-size-fits-all approach results in cluttered navigation, making it difficult for users to find relevant tools or resources efficiently. For instance, students struggle to locate specific study materials, while staff face challenges managing their teaching responsibilities within an unintuitive system. This poor usability compounds the frustration felt across all user groups, diminishing the overall effectiveness of the learning process.

Content management is another critical pain point. The current setup depends on manual processes or disjointed tools like email chains and shared drives to distribute educational resources such as lecture notes, videos, or assignments. This fragmentation leads to delays in content availability, misplaced files, and inconsistent access for students, particularly those in remote areas or with limited technological resources. The absence of a centralized system exacerbates these issues, leaving both staff and students at a disadvantage when timely access to materials is essential for academic success.

Furthermore, the lack of real-time monitoring tools hampers administrators and staff alike. Without detailed statistics on user activity, content usage, or staff performance, it becomes nearly impossible to assess engagement levels or identify areas needing improvement. This blind spot prevents the college from optimizing its educational strategies or addressing underperformance proactively. The problem is particularly acute in

Accessibility also remains a significant barrier. Many existing systems are not optimized for diverse devices, such as smartphones or tablets, which are increasingly common among students. This lack of responsiveness limits the reach of educational resources, especially for those who rely on mobile technology rather than desktop computers. Combined with the aforementioned issues, these shortcomings create a disjointed and inefficient learning experience that fails to meet modern expectations.

The impact of these challenges is profound: students disengage due to lack of interaction and poor resource access, staff waste time navigating inefficient systems, and administrators cannot effectively oversee or enhance the educational process. "LEARNING SPARK" tackles these problems head-on by introducing a web-based LMS designed specifically for Don Bosco College. Built with PHP, MySQL, HTML, CSS, and JavaScript, it offers a robust solution tailored to the institution's needs.

Key features include real-time chat functionality to enable instant communication between students and staff, breaking down the barriers of static learning. Personalized pages for each user role—Admin, Staff, and Users—ensure a streamlined and intuitive experience, reducing navigation frustrations. A centralized content management system simplifies the upload, organization, and distribution of educational materials, ensuring all students have equal and timely access. Detailed analytics provide administrators and staff with insights into engagement and performance, empowering data-driven decisions. Finally, the platform's responsive design guarantees accessibility across devices, from desktops to mobile phones, catering to the diverse technological realities of the student body. By addressing interactivity, usability, content management, monitoring, and accessibility, "LEARNING SPARK" transforms the digital learning experience at Don Bosco College, bridging the gap between traditional education and the demands of a modern, post-COVID academic environment.

1.2.2.1 Objectives of the Project:

- Enhance Real-Time Interaction Between Students and Educators
- The primary objective is to foster dynamic engagement by integrating real-time chat functionality into the platform.
- This allows students to ask questions and receive immediate feedback from staff, overcoming the limitations of static content.
- By enabling instant communication, "LEARNING SPARK" aims to boost student motivation and comprehension of course material.

The goal is to create a collaborative learning environment that mirrors the interactivity

Of the classes .

- Provide Personalized and Role-Specific User Interfaces
- "LEARNING SPARK" seeks to improve usability by offering tailored pages for Admin, Staff, and User roles.
- This ensures that each user group—administrators, educators, and students—accesses tools and resources relevant to their needs.
- By eliminating generic, cluttered interfaces, the system reduces navigation difficulties and enhances efficiency.
- The objective is to deliver a seamless, intuitive experience that caters to the distinct responsibilities

1.2.2.2 Available Input:

○ User Credentials:

- User ID
- Password
- Role (Admin/Student/Staff)

○ Resource Materials:

- Title
- Description
- Resource Type

1.2.2.3 Expected Output:

○ User Authentication & Access Control:

- Password encryption and authentication via MySQL for security.
- Role-based access (Admin/Staff/Student).

○ Admin Functionalities:

- Manage users (Add, Update, Delete).
- Post, update, and delete course details.
- Add, update, and delete assignment details.
- Upload and manage learning resources.

○ Student Functionalities:

- View available courses.
- Access learning resources.

○ Staff Functionalities:

- View assigned courses.
- Upload assignment details.
- Post blogs and resources.
- Chat with students and admin.

Process Logic:

○ User Login:

- Authenticate users using MySQL authentication.
- Redirect users based on their roles (Admin/Staff/Student).

○ Admin Dashboard:

- Manage courses (Post, Edit, Delete).
- Manage assignments (Add, Edit, Delete).
- Manage learning resources (Upload, Update, Delete).

○ User Management:

- Add new users with email authentication.
- Assign roles (Admin/Staff/Student).
- Update user information.
- Remove users if necessary.

1.2.2.4 Stakeholder:

- Prof. John Benedict,
- Don Bosco College, Yelagiri Hills.

1.2.2.5 End Users:

○ Students:

- Seeking course materials, assignments, and educational resources.
- Preparing for academic assessments and skill development.

○ Admin Users:

- Faculty or administrators managing courses and student progress and the PHD completed prefacers.

1.2.3 Proposed Solutions: System Functionalities:

The proposed system will have the following functionalities to address the problem described above:

○ Secure Login:

- MySQL authentication for students, staff, and admins.
- Password encryption and role-based access control.

○ Admin Panel:

Full CRUD (Create, Read, Update, Delete) operations for managing:

- Users (Admins, Staff, & Students)
- Course Listings
- Assignments
- Study Resources

○ Course Posting:

- Admins can post courses with complete details.
- Set duration and enrollment links.
- Search and filter courses based on title, eligibility, and mode.
- Notification alerts for new course postings.

Project Features:

○ MySQL Integration:

- MySQL database for real-time data storage.
- MySQL authentication for secure logins.
- Server storage for hosting resources and submissions.

○ Real-time Updates:

- Automatic syncing of course postings and assignment schedules.
- Push notifications for course updates and assignment reminders.

○ User-friendly Interface:

- Developed using PHP, HTML, CSS, and JavaScript.
- Clean UI/UX design with intuitive navigation.
- Responsive design and accessibility options.

○ Role-based Access:

- Admins: Manage courses, assignments, users, and resources.
- Staff: Upload assignments, post resources, chat with users.
- Students: View course listings, submit assignments, access resources.

III Hardware Requirements:

Development System:

- Minimum 8GB RAM, Intel i3+ processor.
- (Recommended: 16GB RAM, i5+).
- SSD storage for faster performance.

Testing Devices:

- Web browsers (Chrome, Firefox, Edge; latest versions recommended).
- Mobile devices for responsive testing (iOS/Android).

IV Other Requirements:

- Stable internet connection for MySQL operations and real-time updates.
- Web hosting server with PHP and MySQL support.

1.2.4 Name and Address of the Client:

Name: Prof. John Benedict

Institution: Don Bosco College (Co -ED) Yelagiri Hills

1.2.5 Future Enhancements:

AI-based Course Recommendations:

- Machine learning to suggest courses based on student profiles.
- Personalized course alerts and learning paths.

Assignment Upload & Staff Access:

- Students can upload assignments for staff to review.
- Staff can filter submissions based on course and performance.

Chat Support for Student Queries:

- AI-powered chatbot for instant assistance.
- Live chat feature for real-time staff support.

CHAPTER – II

SYSTEM STUDY

2.1 INTRODUCTION

Introduce Your Organization:

This project is developed for Don Bosco College (Co-Ed), Yelagiri Hills, an institution that facilitates online education and academic development for students. The college aims to provide students with course materials, assignments, and learning resources in a streamlined manner.

"LEARNING SPARK" is designed to eliminate the inefficiencies of manual content delivery and assignment management, ensuring a real-time, digital solution for students, staff, and administrators.

Explain the General System

"LEARNING SPARK" is a web-based system that enables students to connect with educational content and participate in academic assignments.

User Roles:

- View the posting blogs.
- View and download the resource.
- Ask the Realtime query.

Admin:

- Insert , delete, update the students, staff, and admins.
- Reply to the Staff messages.

If an Existing System is Available, Describe It

Currently, the learning management process is managed manually using notices, spreadsheets, and emails. The system is:

Time-consuming due to manual content distribution.

Error-prone, leading to misplaced resources.

Difficult to track, as records are scattered across multiple formats.

Lacks real-time updates, causing delays in Take a notes and ask questions

ii. Explain the Problem

- Delayed content postings: Course materials and blogs are not updated instantly, leading to missed learning opportunities.
- No centralized database: Study materials, blogs, and messages are stored separately, making it hard to track data.
- Lack of authentication: Without a proper system, controlling access is challenging.
- Student difficulties: Students struggle to find relevant staff content and resources.

iii. Suggest the Solution

The proposed "LEARNING SPARK" will provide a centralized, real-time digital system with:

- MySQL Authentication: Secure login for admins, staff, and students.
- Real-time Updates: Blogs and study materials update instantly.
- Cloud Database: All course materials, blogs, and messages are stored in MySQL.
- User-Friendly Interface: Developed using PHP, HTML, CSS, and JavaScript.

2.2 LIST OF MODULES

The application consists of the following key modules:

1. Manage Users
2. Manage Jobs
3. Manage Tests
4. Manage Resources
5. Student Dashboard

Module 1:

Manage Users Description: Allows the admin to manage user accounts (admins, staff, and students).

Reference Document: MySQL Documentation

Activities:

1. User Registration:

- Purpose: Add new users to the system.
- Entry Criteria: Admin clicks "Add User."
- Input: User ID, Name, Email, Password, Role (Admin/Staff/Student).
- Output: User registered successfully.
- Exit Criteria: User data stored in MySQL users table.

2.Update Password:

- Purpose: Modify user credentials.
- Entry Criteria: Admin selects a user and clicks "Update Password."
- Input: New Password.
- Output: Password updated successfully.
- Exit Criteria: Password securely stored in MySQL users table.

Module 2:**Manage Staff Content**

Description: Staff can add, update, and delete study materials (videos or documents).

Reference Document: MySQL Documentation

Activities:**1. Add study materials:**

- Purpose: Register new study materials.
- Entry Criteria: Staff clicks "Add Content."
- Input: Title, Type (Video/Document), File Path, Description.
- Output: Material posted successfully.
- Exit Criteria: Content details stored in MySQL staff_content table.

2. Update/Delete study materials:

- Purpose: Modify or remove study materials.
- Entry Criteria: Staff selects a content item and clicks "Update/Delete."
- Output: Material updated/deleted successfully.
- Exit Criteria: Content list updated in MySQL staff_content table.

Module 3:**Manage Blogs:**

Description: Admin can create, modify, and delete placement test schedules. **Reference Document:** Firebase Database & Authentication

Documentation **Activities:**

1. Add Blog:

- Purpose: Post daily blogs.
- Entry Criteria: Staff clicks "Add Blog."
- Input: Title, Content (Text), Image Path (optional).
- Output: Blog added successfully.
- Exit Criteria: Blog details stored in MySQL staff_blogs table.

2. Update/Delete Blog:

- Purpose: Modify or remove blogs.
- Entry Criteria: Staff selects a blog and clicks "Update/Delete."
- Output: Blog updated/deleted successfully.
- Exit Criteria: Blog list updated in MySQL staff_blogs table.

1. Update/Delete Test:

- **Purpose:** Modify or remove tests.
- **Entry Criteria:** Admin selects a test and clicks "Update/Delete."
- **Output:** Test updated/deleted successfully.
- **Exit Criteria:** Test list updated in Firebase.

Module 4:**Manage Messaging:**

Description: Students can message Staff, and Staff can reply; Admins manage Staff messages. **Reference Document:** MySQL Documentation

Activities:**1. Send Message:**

- Purpose: Facilitate communication between Students and Staff.
- Entry Criteria: Student selects Staff and clicks "Send Message."
- Input: Message Text.
- Output: Message sent successfully.
- Exit Criteria: Message stored in MySQL user_to_staff_messages table.

2. Replay To The Message:

- Purpose: Allow Staff to respond to Student queries.
- Entry Criteria: Staff selects a message and clicks "Reply."
- Input: Staff Reply Text.
- Output: Reply sent successfully.
- Exit Criteria: Reply updated in MySQL user_to_staff_messages table.

Module 5: StudentDashboard

Description: Students can select Staff, view blogs/content, download resources, and message Staff.

Reference Document: MySQL Documentation

Activities:**1.View Staff Content:**

- Purpose: Access Staff-posted blogs and materials.
- Entry Criteria: Student selects a Staff member.
- Input: Staff ID (from user_preferences table).
- Output: Blogs and materials displayed successfully.
- Exit Criteria: Content retrieved from MySQL staff_blogs and staff_content tables.

2. Download Resources:

- Purpose: Download videos, documents, and notes.
- Entry Criteria: Student clicks "Download" on a resource.
- Output: Resource downloaded successfully.
- Exit Criteria: File accessed from MySQL staff_content table's file_path.

1. User Table.

Data Member Name	Description	Type	Additional Info	Default Value	Mandatory?	Unique?
User ID	Unique identifier for user	Integer	Positive integer only	0	Yes	Yes
Name	Full name of the user	Character	Min: 5, Max: 100	Null	Yes	No
Email	User's email	String	Valid email format	Null	Yes	Yes
Password	Login password	String	Encrypted storage	Null	Yes	No
Role	User role (Admin/Staff/User)	String	Enum: Admin, Staff, User	User	Yes	No

2. Staff Content Table.

Data Member Name	Description	Type	Additional Info	Default Value	Mandatory?	Unique?
Content ID	Unique identifier for content	Integer	Positive integer only	0	Yes	Yes
Staff ID	ID of staff posting content	Integer	References users table	Null	Yes	No
Title	Name of the content	Character	Min: 5, Max: 100	Null	Yes	No
Type	Content type (Video/Document)	String	Enum: Video, Document	Null	Yes	No
File Path	Path to content file	String	Max: 255, Valid path	Null	Yes	No

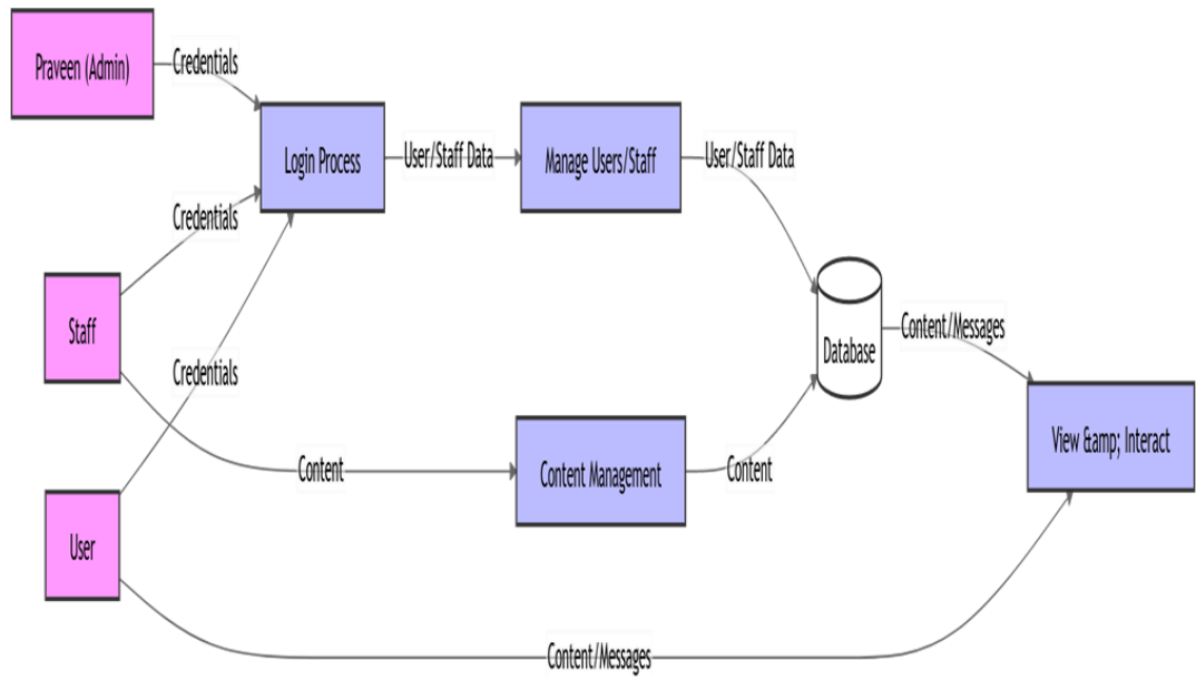
3.Staff Blogs Table:

Data Member Name	Description	Type	Additional Info	Default Value	Mandatory?	Unique?
Blog ID	Unique identifier for blog	Integer	Positive integer only	0	Yes	Yes
Staff ID	ID of staff posting blog	Integer	References users table	Null	Yes	No
Title	Name of the blog	Character	Min: 5, Max: 100	Null	Yes	No
Content	Blog text content	String	Min: 10, Max: 500	Null	Yes	No
Image Path	Path to blog image (optional)	String	Max: 255, Valid path	Null	No	No

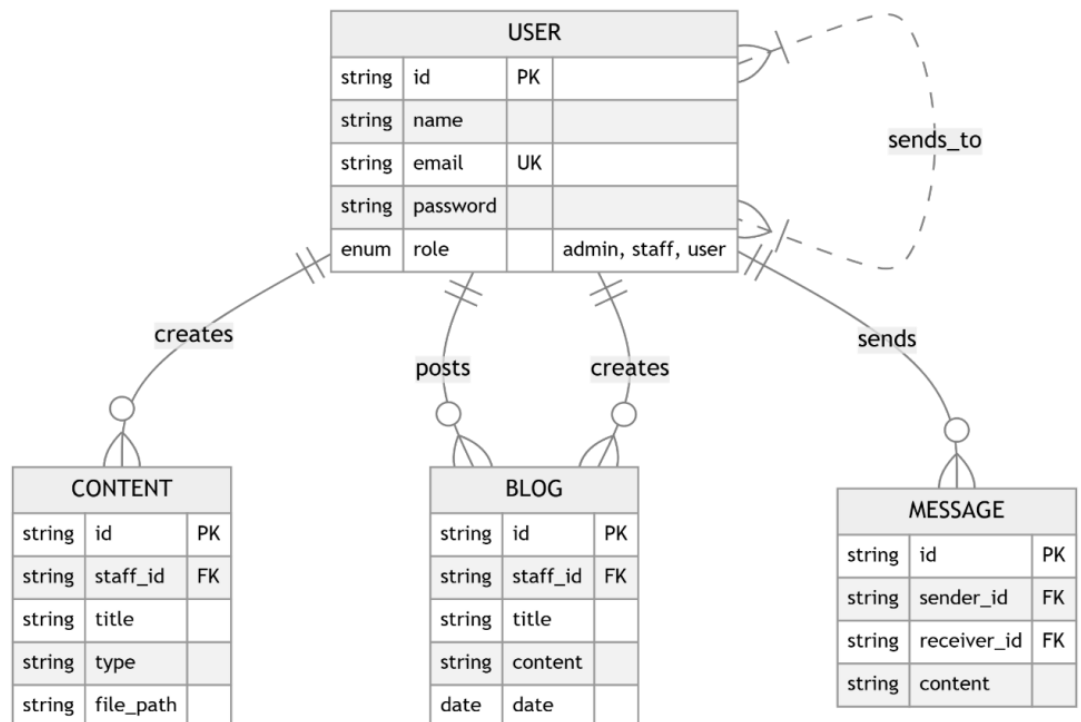
4.User to Staff Messages Table:

Data Member Name	Description	Type	Additional Info	Default Value	Mandatory?	Unique?
Message ID	Unique identifier for message	Integer	Positive integer only	0	Yes	Yes
User ID	ID of student sending message	Integer	References users table	Null	Yes	No
Staff ID	ID of staff receiving message	Integer	References users table	Null	Yes	No
Message	Message text	String	Min: 1, Max: 500	Null	Yes	No
Staff Reply	Reply from staff (optional)	String	Max: 500	Null	No	No

2.3 DATA FLOW DIAGRAM



2.4 ER-DIAGRAM



CHAPTER – III

SOFTWARE REQUIREMENTS

3.1 VISION DOCUMENT

3.1.1 Introduction

"LEARNING SPARK" is designed to streamline the learning management process for Don Bosco College (Co-Ed), Yelagiri Hills by providing a centralized platform where students can access staff-posted educational content, engage with daily blogs, and communicate with staff for academic support.

3.1.2 Business Objectives

- To digitize the learning management process for students, staff, and administrators.
- To eliminate manual errors and improve resource accessibility.
- To provide secure authentication for students, staff, and admins.
- To offer real-time updates for blogs, study materials, and messages.
- To enhance the academic experience of students by connecting them with staff content.

3.1.3 Key Features

- User Authentication – Secure login using MySQL Authentication.
- Admin Dashboard – Manage users, staff content, blogs, and messages.
- Content Listings – Post, update, and delete study materials (videos/documents).
- Blog Management – Schedule and modify daily blogs (text/images).
- Messaging System – Facilitate student-staff communication with replies.

3.1.4 Future Scope

- AI-based content recommendations based on student preferences.
- Note-taking and download feature for study materials.
- Integration with staff for personalized learning paths.
- Live chat support for student queries.

3.1 SUPPLEMENTARY SPECIFICATION

3.1.1 Introduction

This document provides a comprehensive overview of the Supplementary Specification for "LEARNING SPARK." It captures non-functional requirements such as security, usability, reliability, performance, supportability, legal compliance, and system constraints that are not covered in the use-case model.

Purpose

The purpose of this document is to define the non-functional requirements of "LEARNING SPARK," **ensuring:**

- Usability – The system should be easy to navigate for students, staff, and administrators.
- Reliability – The system should operate with minimal downtime.
- Performance – The system should efficiently handle a large number of users.
- Supportability – The system should support future enhancements.

3.1.2 Scope

This Supplementary Specification applies to "LEARNING SPARK," which facilitates staff content posting, blog management, student-staff messaging, and resource access. It is used by:

- Administrators – Managing users, staff content, blogs, and messages.
- Staff – Posting blogs, uploading materials, and replying to students.
- Students – Selecting staff, accessing content, and messaging staff.

3.1.3 Definitions, Acronyms, and Abbreviations

- Admin – The system administrator responsible for managing users, content, blogs, and messages.
- MySQL – A relational database management system used for data storage.
- UI/UX – User Interface/User Experience, ensuring a user-friendly system.
- PHP – A server-side scripting language used for web development and database interactions.

3.1.4 References

- PHP Official Documentation
- MySQL Documentation
- HTML/CSS/JavaScript Developer Guides
- Bootstrap UI/UX Guidelines

This document outlines system functionalities, usability, reliability, performance, security measures, and other constraints affecting development.

3.1.1 Functionality

The system provides role-based access with the following functionalities:

3.1.1.1 Admin Functionalities

User Management – Add, update, and delete student/staff/admin accounts. Content Management – Post, update, and delete study materials. Messaging Oversight – Monitor and manage staff messages.

3.1.1.2 Staff Functionalities

- Post Content – Upload study materials (videos/documents) for student access.
- Post Blogs – Share daily blogs (text/images) with students.
- Reply to Messages – Respond to student queries via messaging system.

3.1.1.3 Student Functionalities

- View Content – Browse staff-posted materials and download resources.
- View Blogs – Access daily blogs posted by selected staff.
- Send Messages – Communicate with staff and receive replies.

3.1.2 Usability

3.1.2.1 Usability Requirements

- User-friendly Interface – The system must be designed with Bootstrap UI principles.
- Minimal Training – Users should be able to navigate without formal training.
- Responsive Design – The app must be optimized for both desktop and mobile devices.
- Accessibility Features – Support for screen readers and high-contrast themes.

3.1.2.2 Reliability Requirements

- 99.5% Uptime Guarantee – The system must be available with minimal downtime.
- Data Consistency – MySQL should ensure real-time data synchronization across devices.
- Automatic Backup – Data should be backed up regularly to prevent loss.
- Failure Recovery – System should automatically restart services in case of crash.

3.1.2.3 Performance Requirements

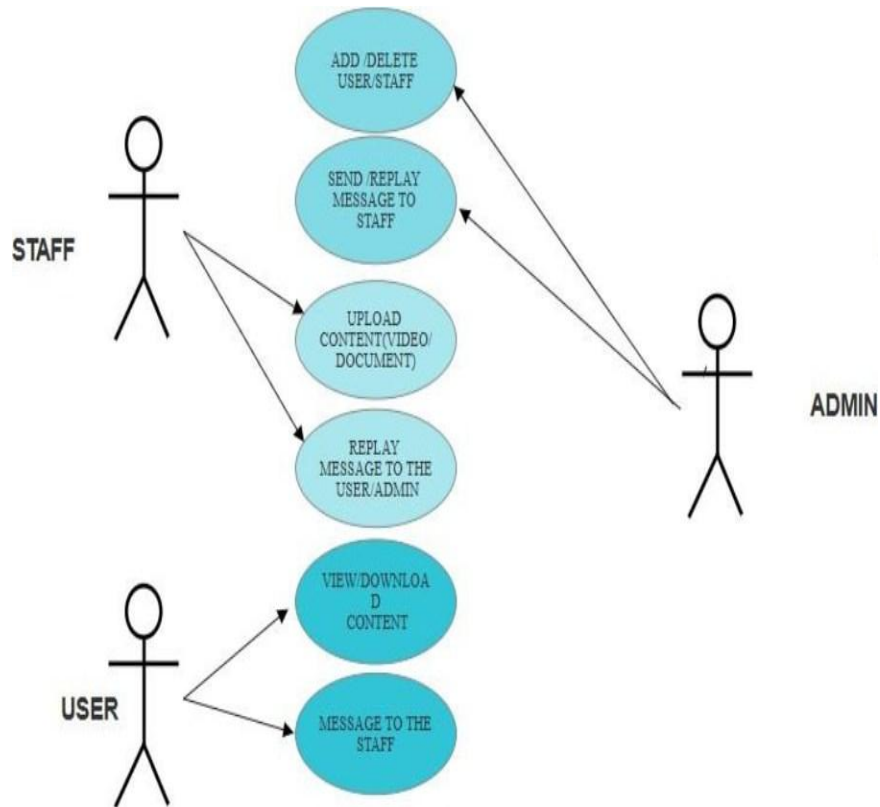
- Fast Response Time – Content searches should load within 2 seconds.
- High Concurrency – The system must support at least 1000 simultaneous users.
- Optimized Database Queries – MySQL queries should be structured to reduce latency.
- Efficient Image Loading – Use lazy loading for blog and content images.

3.1 USE CASE DOCUMENT

While seeing just the static idea of framework is inadequate, Use-Case graphs assists with giving the dynamic perspective on the framework. Use Case graphs models the framework and the subsystems of "LEARNING SPARK." There are some outer and interior factors that mark the dynamic idea of the Use Case chart. The point of use case diagrams is mainly to identify what happens inside / outside of your well defined system boundaries. Actors are elements that exist outside your system boundaries and either trigger actions provided by your system, or being triggered by your system to do something. The Actors can communicate with "LEARNING SPARK" actively or passively. That can be expressed with directed associations in UML mainly to identify

what happens inside / outside of your well defined system boundaries. Actors are elements that exist outside your system boundaries and either trigger actions provided by.

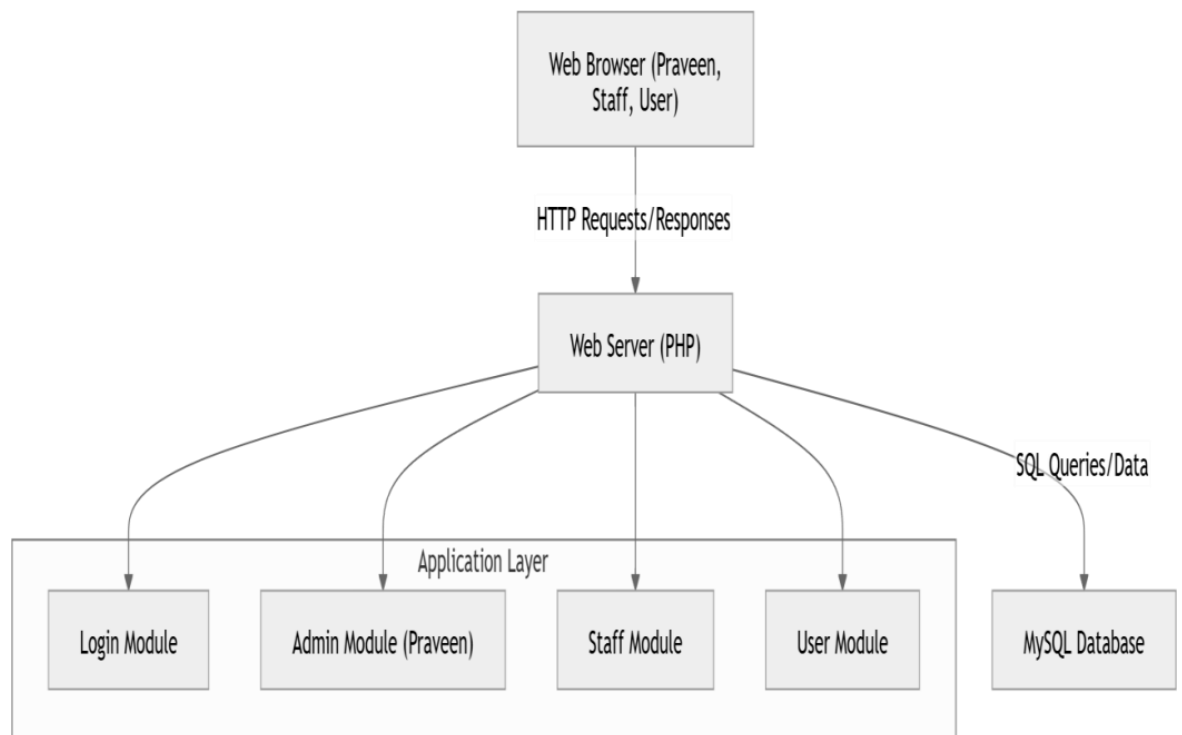
Use Case Diagram



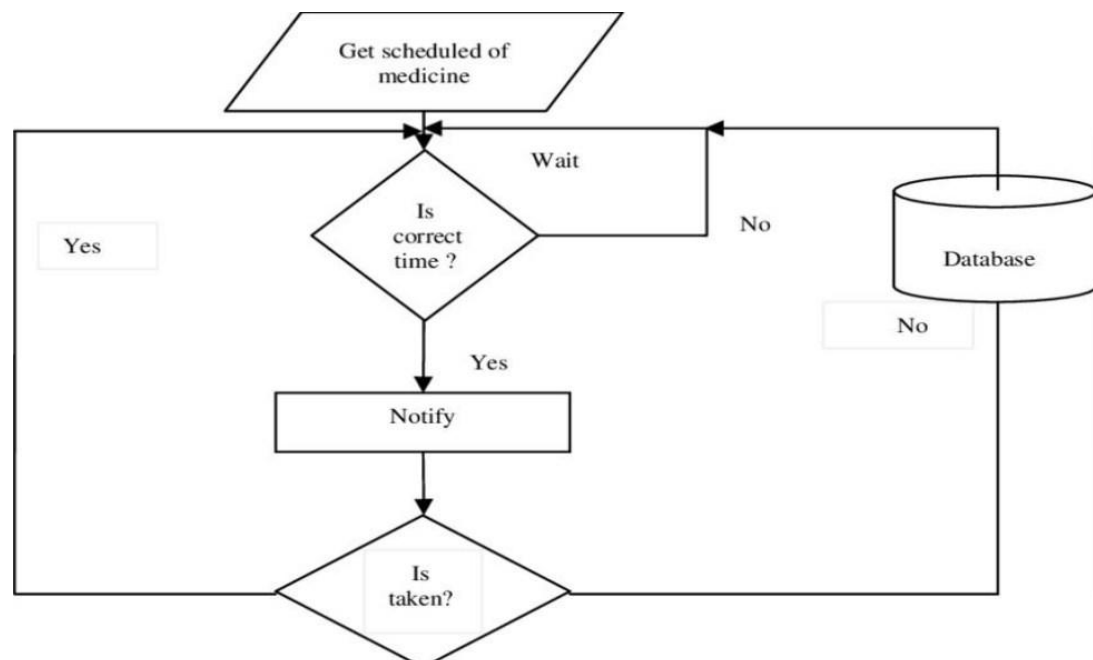
CHAPTER – IV

SYSTEM DESIGN

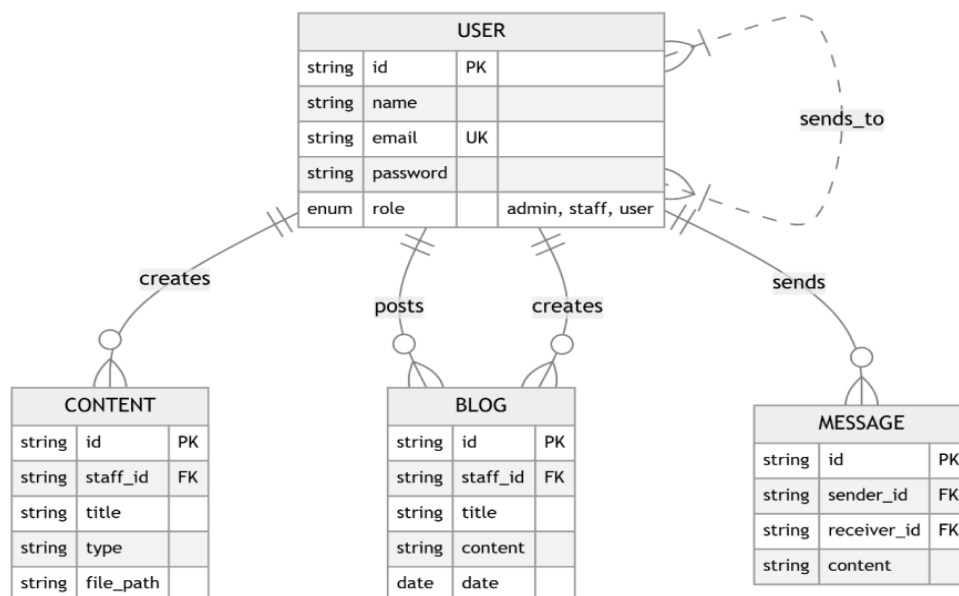
1.1 ARCHITECTURAL DIAGRAM



4.2. PROCESS DESIGN



4.3 DATABASE DESIGN

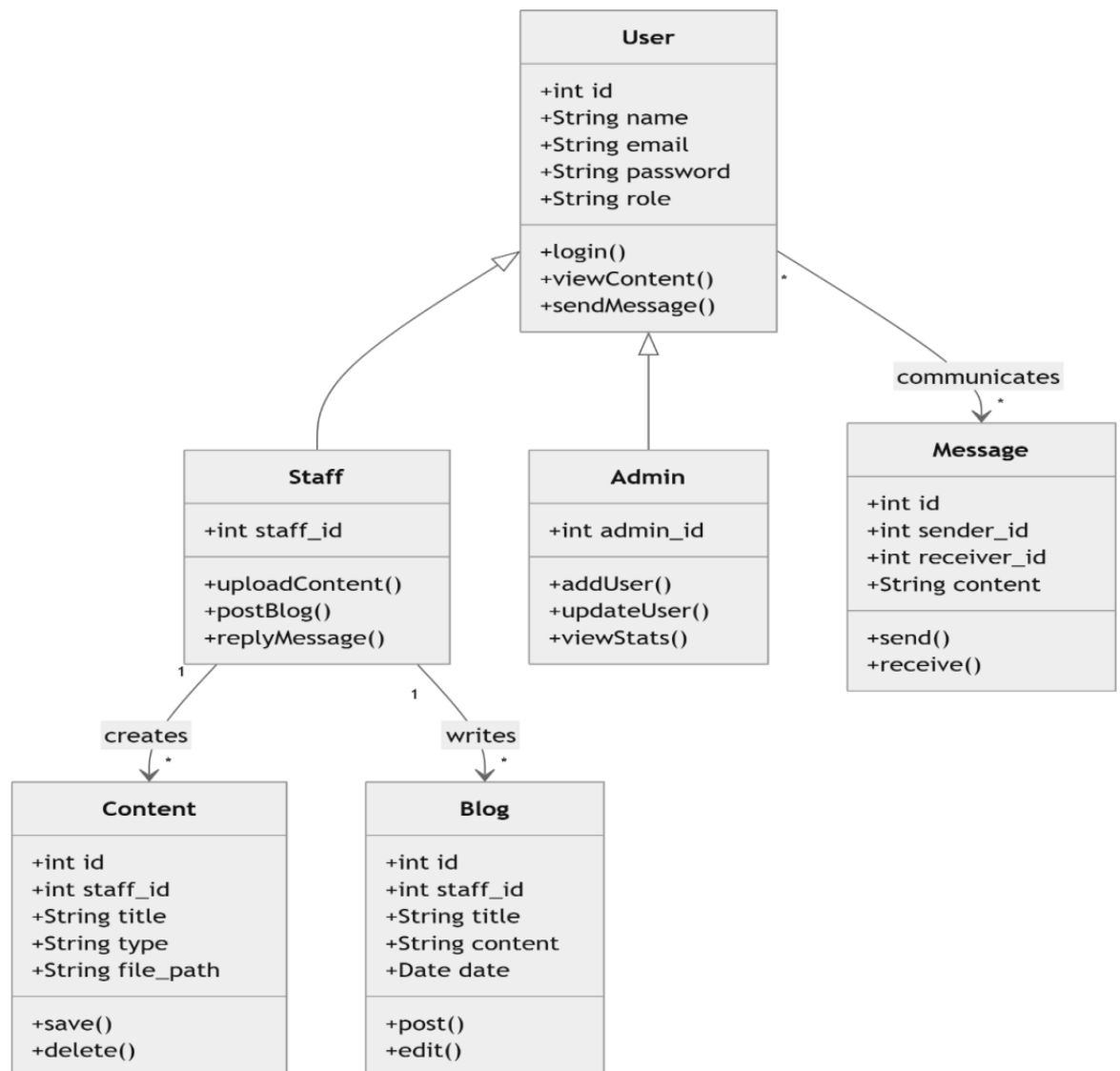


4.3.1 DATABASE TABLE

S.no	Field	Type	Width	Constraints	Description
1	id	Int	11	Primary Key	User ID
2	name	Varchar	50	-	User Name
3	email	Varchar	100	Unique	Email Address
4	password	Varchar	255	-	Password
5	role	Enum	10	-	Role (user/staff/admin)

4.4 CLASS DESIGN

Class Diagram



4.3 FORM DESIGN

Login Page:

LEARNING SPARK

LEARNING SPARK

Email Address

Enter your email

Password

Enter your password

Login Now

New to the portal? [Register here](#)

Register Page:

LEARNING SPARK

LEARNING SPARK

Full Name

Enter your full name

Email Address

Enter your email

Password

Create a password

Role

Select Role

Register Now

Already a member? [Sign in here](#)

Admin Page:

Manage Users & Staff

Name	Email	Role	Actions
PRASANNA	pricy@gmail.com	User	Save Cancel Delete
PRAVEEN RAJ D	praveenrajnbooco@gmail.com	User	Save Cancel Delete
RONALD	rg@gmail.com	Staff	Save Cancel Delete
RUSO	ruso@gmail.com	Staff	Save Cancel Delete
SANJAY	sanjay@gmail.com	Staff	Save Cancel Delete

Add New User/Staff

Admin Profile:

Admin Profile

Manage your account details

Full Name

Email Address

Password

[Update Profile](#)

Admin Notification:

Notifications

Manage your messages

Messages from Staff

No messages from staff at this time.

Your Replies to Staff

No replies sent to staff yet.

Staff Profile:



Manage Profile

Profile updated successfully!



Name

Email

Profile Photo

 No file chosen

[Update Profile](#)

Staff Upload Content:

Upload Content

Title

Type

Video

File

Choose File No file chosen

Upload

Staff content Preview:

Content Preview	
GOOD EVENING (blog) - 2025-03-18 14:36:57	View Delete Update
PROJECT DOCUMENT (document) - 2025-03-18 14:34:50	View Delete Update

Staff Post Blogs:

Post Blog

Title

Content

Blog Image (optional)

Choose File No file chosen

Post Blog

Staff View Status:

	Content Uploaded	Total Blogs Posted	Total Admin Messages
	2	0	0
			
			
			
			
			
			
			

Staff Messages:

Send Message to Admin

Message

Send

Messages with Admin

No messages with admin yet.

Messages from Users

User: please upload this document in pdf

You: yes i will make that

Delete

2025-03-18 14:42:58

User Profile Page:

LEARNING SPARK

Your Profile

Profile updated successfully!



Name: PRASANNA
Email: pricy@gmail.com

Edit Profile

Full Name

Email Address

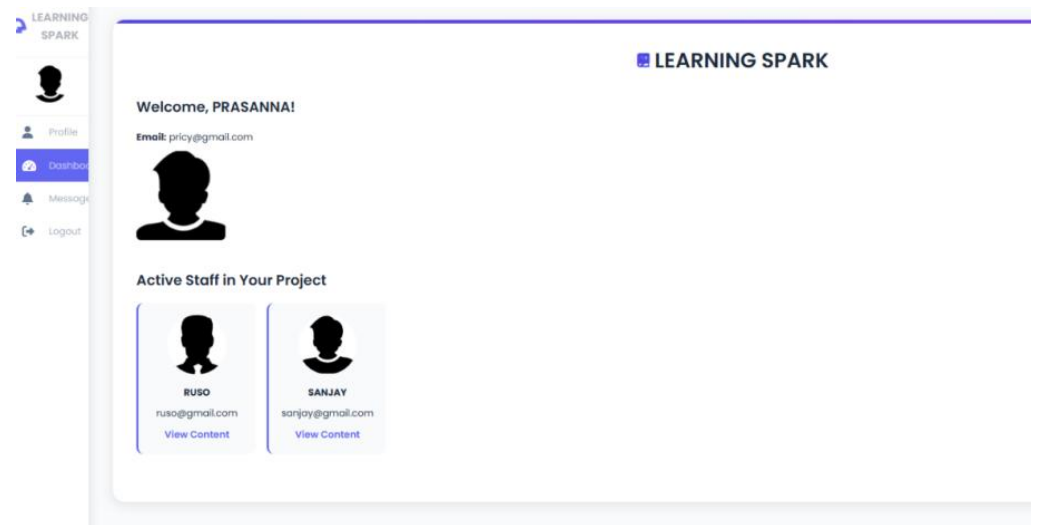
New Password (leave blank to keep current)

Profile Photo

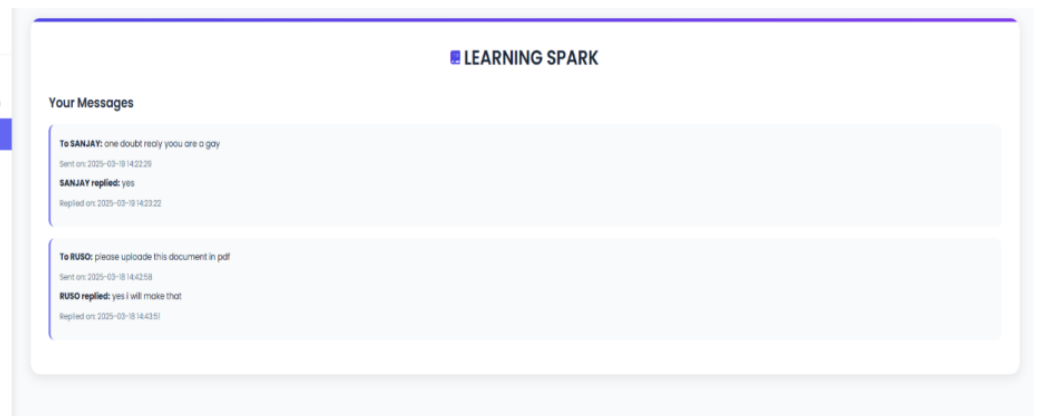
Choose File No file chosen

Update Profile

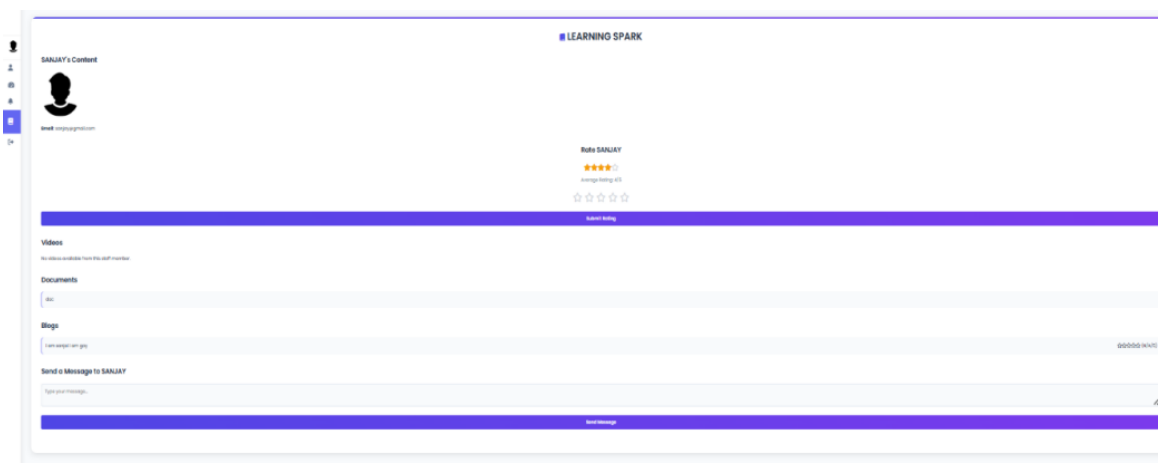
User Dashboard Page:



User Message Page:



User Content Page:



CHAPTER – V

SOURCE CODE

1. SOURCE

CODE 1.1 Login

Page

```
<?php session_start();

// Database connection
$conn = new mysqli("localhost", "root", "", "learning"); if ($conn->connect_error) {
die("Connection failed: " . $conn->connect_error);
}
// Handle login
if (isset($_POST['login'])) {
$email = $conn->real_escape_string($_POST['email']);
$password = $conn->real_escape_string($_POST['password']); Plain text
password
$sql = "SELECT * FROM users WHERE email = '$email' AND password = '$password'";
$result = $conn->query($sql);
if ($result->num_rows > 0) {
$user = $result->fetch_assoc();
$_SESSION['role'] = $user['role'];
$_SESSION['email'] = $user['email'];
$_SESSION['name'] = $user['name'];
// Redirect based on role
if ($user['role'] === 'admin') { header("Location: admin_dashboard.php");
} elseif ($user['role'] === 'staff') {
header("Location: staff.php"); // Adjust if staff.php doesn't exist
} else { // 'user' header("Location: user.php");
exit();
} else {
echo "<script>alert('Invalid credentials!');</script>";
}
}
$conn->close();
?>
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>LMS - Login</title>
<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.0.0-beta3/css/all.min.css">
<style>
* { margin: 0;
padding: 0;
box-sizing: border-box;
font-family: 'Arial', sans-serif;
}
body {
background: linear-gradient(135deg, #1e3c72, #2a5298, #3b5998, #8e44ad); background-size:
300% 300%;
animation: gradientShift 20s ease infinite; color: #fff;
```

```

min-height: 100vh; display: flex;
justify-content: center; align-items: center; overflow: hidden;

}
@keyframes gradientShift {
0% { background-position: 0% 0%; }
50% { background-position: 100% 100%; }
100% { background-position: 0% 0%; }
}
.container {
background: rgba(255, 255, 255, 0.1); padding: 40px;
border-radius: 20px;
box-shadow: 0 0 40px rgba(0, 0, 0, 0.5); backdrop-filter: blur(20px);
width: 100%;
max-width: 450px;
animation: pulseContainer 3s infinite ease-in-out;
}
@keyframes pulseContainer {
0% { transform: scale(1); box-shadow: 0 0 20px rgba(255, 255, 255, 0.3); }
50% { transform: scale(1.01); box-shadow: 0 0 40px rgba(255, 255, 255, 0.5); }
100% { transform: scale(1); box-shadow: 0 0 20px rgba(255, 255, 255, 0.3); }
}
h1 {
text-align: center; margin-bottom: 30px; font-size: 2.5em;
text-shadow: 0 0 15px rgba(255, 255, 255, 0.8); letter-spacing: 1px;
}
input, button { width: 100%; padding: 12px; margin: 12px 0; border: none; border-radius: 10px;

background: rgba(255, 255, 255, 0.25); color: #fff;
font-size: 1em; transition: all 0.4s ease;
}
input:focus { outline: none;
background: rgba(255, 255, 255, 0.4);
box-shadow: 0 0 10px rgba(255, 255, 255, 0.7);
}
button {
background: #00ddeb; cursor: pointer;
font-weight: bold; color: #1e3c72;
}
button:hover { background: #3b5998;
transform: translateY(-3px);
box-shadow: 0 5px 20px rgba(59, 89, 152, 0.7);
}
.link {
text-align: center; margin-top: 20px;
}
.link a {
color: #00ddeb;
text-decoration: none; font-weight: bold;
}
.link a:hover { color: #ffeb3b;
}
.particles {

position: absolute; width: 100%;
height: 100%; pointer-events: none; overflow: hidden;
}
.particle { position: absolute;

```

```

background: rgba(255, 255, 255, 0.7);
border-radius: 50%; animation: float 10s infinite;
}
@keyframes float {
0% { transform: translateY(100vh) scale(0.5); opacity: 0.7; } 100% { transform: translateY(-10vh)
scale(1.2); opacity: 0; }
}
</style>
</head>
<body>
<div class="particles" id="particles"></div>
<div class="container">
<h1>Login</h1>
<form method="POST">
<input type="email" name="email" placeholder="Email" required>
<input type="password" name="password" placeholder="Password" required>
<button type="submit" name="login">Login</button>
</form>
<div class="link">
<p>Not registered? <a href="register.php">Register here</a></p>
</div>
</div>
<script>
const particlesContainer = document.getElementById('particles');

for (let i = 0; i < 80; i++) {
const particle = document.createElement('div'); particle.classList.add('particle');
particle.style.width = `${Math.random() * 5 + 2}px`; particle.style.height = particle.style.width;
particle.style.left = `${Math.random() * 100}vw`;
particle.style.animationDuration = `${Math.random() * 6 + 4}s`; particle.style.animationDelay =
`${Math.random() * 5}s`; particlesContainer.appendChild(particle);
}
</script>
</body>
</html>

```

1.1 Admin Dashboard:

```

<?php session_start();
// Check if admin is logged in
if (!isset($_SESSION['role']) || $_SESSION['role'] !== 'admin') { header("Location:
login.php");
exit();
}
// Ensure email is set in session
if (!isset($_SESSION['email']) || empty($_SESSION['email'])) { die("Session email not set
or empty.");
}
// Database connection
$conn = new mysqli("localhost", "root", "", "learning"); if ($conn->connect_error) {
die("Connection failed: " . $conn->connect_error);
}
// Fetch stats
$result = $conn->query("SELECT COUNT(*) as count FROM users WHERE role
= 'user'");
if ($result === false) die("Query failed (total_users): " . $conn->error);

```



```

$total_users = $result->fetch_assoc()['count'];

$result = $conn->query("SELECT COUNT(*) as count FROM users WHERE role
= 'staff'");
if ($result === false) die("Query failed (total_staff): " . $conn->error);
$total_staff = $result->fetch_assoc()['count'];
// Fetch all users and staff
$users = $conn->query("SELECT id, name, email, role FROM users WHERE role IN
('user', 'staff') ORDER BY role, name");
if ($users === false) die("Query failed (users): " . $conn->error);
// Fetch admin ID
$stmt = $conn->prepare("SELECT id FROM users WHERE email = ?"); if ($stmt ===
false) die("Prepare failed (admin_id): " . $conn->error);
$stmt->bind_param("s", $_SESSION['email']);
$stmt->execute();
$result = $stmt->get_result();
$row = $result->fetch_assoc(); if ($row === null) {
die("No user found with email: " . htmlspecialchars($_SESSION['email']));
}
$admin_id = $row['id'];
$stmt->close();
// Fetch unread notification count
$stmt = $conn->prepare("SELECT COUNT(*) as count FROM
staff_to_admin_messages WHERE admin_id = ? AND is_read = 0");
if ($stmt === false) die("Prepare failed (unread_count): " . $conn->error);
$stmt->bind_param("i", $admin_id);
$stmt->execute();
$result = $stmt->get_result();
$unread_count = $result->fetch_assoc()['count'];
$stmt->close();
// Handle delete
if (isset($_POST['confirm_delete'])) {
$id = intval($_POST['delete_id']); // Use intval for safety
// Start a transaction to ensure data consistency
$conn->begin_transaction();
try {
// Check if the user is a staff member
$stmt = $conn->prepare("SELECT role, profile_photo FROM users WHERE id = ? AND
role != 'admin'");
$stmt->bind_param("i", $id);
$stmt->execute();
$result = $stmt->get_result();
$user = $result->fetch_assoc();
$stmt->close();
if ($user) {
if ($user['role'] === 'staff') {
// Delete staff content files and database records
$stmt = $conn->prepare("SELECT file_path, image_path FROM staff_content WHERE
staff_id = ?");
$stmt->bind_param("i", $id);
$stmt->execute();
$content_result = $stmt->get_result();

```

```

while ($content = $content_result->fetch_assoc()) {
// Delete file_path if it exists
if ($content['file_path'] && file_exists($content['file_path'])) {
unlink($content['file_path']);
}
// Delete image_path if it exists (for blogs)
if ($content['image_path'] && file_exists($content['image_path'])) {
unlink($content['image_path']);
}
}
$stmt->close();
// Delete staff content records from the database
$stmt = $conn->prepare("DELETE FROM staff_content WHERE staff_id = ?");
$stmt->bind_param("i", $id);
$stmt->execute();
$stmt->close();

// Delete staff profile photo
if ($user['profile_photo'] && file_exists($user['profile_photo'])) {
unlink($user['profile_photo']);
}
}

// Delete related records from dependent tables
$stmt = $conn->prepare("DELETE FROM blog_ratings WHERE user_id = ?");
$stmt->bind_param("i", $id);
$stmt->execute();
$stmt->close();
$stmt = $conn->prepare("DELETE FROM staff_ratings WHERE user_id = ?");
$stmt->bind_param("i", $id);
$stmt->execute();
$stmt->close();
$stmt = $conn->prepare("DELETE FROM user_to_staff_messages WHERE user_id
= ?");
$stmt->bind_param("i", $id);
$stmt->execute();
$stmt->close();
// Delete the user
$stmt = $conn->prepare("DELETE FROM users WHERE id = ? AND role != 'admin'");
$stmt->bind_param("i", $id);
$stmt->execute();
$stmt->close();

// Commit the transaction
$conn->commit();
echo "<script>alert('User and related data deleted successfully!');
window.location.href='admin_dashboard.php';</script>";
} else {
throw new Exception("User not found or is an admin.");
}
} catch (Exception $e) {

```

```

// Roll back the transaction on error
$conn->rollback();
echo "<script>alert('Error deleting user: " . addslashes($e->getMessage()) . "');
window.location.href='admin_dashboard.php';</script>";
}
}
// Handle update
if (isset($_POST['update_user'])) {
$id = intval($_POST['id']);
$name = $conn->real_escape_string($_POST['name']);
$email = $conn->real_escape_string($_POST['email']);
$role = $conn->real_escape_string($_POST['role']);
$stmt = $conn->prepare("UPDATE users SET name = ?, email = ?, role = ? WHERE id =
? AND role != 'admin'");
$stmt->bind_param("sssi", $name, $email, $role, $id);
$stmt->execute();
$stmt->close();
echo "<script>alert('User updated successfully!');
window.location.href='admin_dashboard.php';</script>";
}
// Handle add new user/staff
if (isset($_POST['add_user'])) {
$name = $conn->real_escape_string($_POST['name']);
$email = $conn->real_escape_string($_POST['email']);
$password = $conn->real_escape_string($_POST['password']); // No hashing, as requested
$role = $conn->real_escape_string($_POST['role']);
$stmt = $conn->prepare("SELECT email FROM users WHERE email = ?");
$stmt->bind_param("s", $email);
$stmt->execute();
$result = $stmt->get_result(); if ($result->num_rows > 0) {
echo "<script>alert('Email already exists!');</script>";
} else {

$stmt = $conn->prepare("INSERT INTO users (name, email, password, role) VALUES (?,
?, ?, ?)");
$stmt->bind_param("ssss", $name, $email, $password, $role);
$stmt->execute();
$stmt->close();
echo "<script>alert('New user added successfully!');
.location.href='admin_dashboard.php';</script>";
}
}
$conn->close();
?>
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Admin Dashboard - Learning Portal</title>
<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.0.0-beta3/css/all.min.css">
<link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600;700&

```

```

display=swap" rel="stylesheet">
<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
padding: 15px; font-weight: 600; color: var(--text);
border-bottom: 2px solid #e2e8f0;
}
.user-table td { padding: 15px;
border-bottom: 1px solid #e2e8f0;
}
.user-table tr:hover { background: #f1f5f9;
}
.user-table input,
.user-table select { width: 100%; padding: 8px;
border: 2px solid #e2e8f0; border-radius: 6px;
font-size: 0.9em; background: #f8fafc; transition: all 0.3s ease;
}
.user-table input:focus,
.user-table select:focus { border-color: var(--primary);
box-shadow: 0 0 5px rgba(99, 102, 241, 0.2); outline: none;
}
.btn {
padding: 10px 20px; border: none;
border-radius: 6px; font-size: 0.9em; font-weight: 600; cursor: pointer;
transition: all 0.3s ease;
}
.save-btn {
background: var(--gradient); color: white;
}
.save-btn:hover { transform: translateY(-2px);
box-shadow: 0 4px 10px rgba(99, 102, 241, 0.3);
}
.cancel-btn { background: #e2e8f0; color: var(--text); margin-left: 10px;
}
.cancel-btn:hover { background: #d1d5db;
}
.delete-btn { background: #ef4444; color: white;
margin-left: 10px;
}
.delete-btn:hover { background: #dc2626;
box-shadow: 0 4px 10px rgba(239, 68, 68, 0.3);
}
.add-btn {
display: inline-block; background: var(--gradient); color: white;
padding: 12px 24px; text-decoration: none;
}
.add-btn:hover {
transform: translateY(-2px);
box-shadow: 0 4px 15px rgba(99, 102, 241, 0.3);
}
.modal { display: none; position: fixed; top: 0;
left: 0;
width: 100%;
height: 100%;

```

```

background: rgba(0, 0, 0, 0.5); justify-content: center;
align-items: center; z-index: 1000;
}
.modal-content { background: var(--card-bg);
padding: 30px; border-radius: 12px;
box-shadow: 0 4px 20px var(--shadow); width: 400px;
text-align: center;
}
.modal-content h2 { font-size: 1.5em; font-weight: 600; color: var(--text);
margin-bottom: 20px;
}
.modal-content .form-group { margin-bottom: 20px;
text-align: left;
}
.modal-content label { display: block;
font-weight: 500; color: var(--text); margin-bottom: 5px;
}
.modal-content input,
.modal-content select { width: 100%;
padding: 10px;
border: 2px solid #e2e8f0; border-radius: 6px;
font-size: 1em;
}
.modal-content input:focus,
.modal-content select:focus { border-color: var(--primary);
box-shadow: 0 0 5px rgba(99, 102, 241, 0.2); outline: none;
}
canvas {
max-width: 100%;
margin: 20px 0;
}
/* Responsive Design */ @media (max-width: 768px) {
.sidebar { width: 60px;
}
.sidebar:hover { width: 200px;
}
.main-content { margin-left: 60px; padding: 20px;
}
.sidebar:hover ~ .main-content { margin-left: 200px;
.stats-grid {

        grid-template-columns: 1fr;

    }
}
</style>
</head>
<body>
<div class="sidebar">
<div class="logo">
<i class="fas fa-graduation-cap"></i>
</div>
<a href="admin_profile.php"><i class="fas fa-user"></i><span>Profile</span></a>

```

```

<a href="admin_dashboard.php" class="active"><i class="fas fa-tachometer-
alt"></i><span>Dashboard</span></a>
<a href="admin_notifications.php" class="notification-bell"><i class="fas fa-
bell"></i><span>Notifications</span></a>
<a href="admin_logout.php"><i class="fas fa-sign-out-
alt"></i><span>Logout</span></a>
</div>
<div class="main-content">
<div class="dashboard-card">
<div class="dashboard-header">
<h1>Admin Dashboard</h1>
<p>Overview of users and staff</p>
</div>
<div class="stats-grid">
<div class="stat-box">
<i class="fas fa-users"></i>
<p>Total Users</p>
<span><?php echo $total_users; ?></span>
</div>
<div class="stat-box">
<i class="fas fa-user-tie"></i>
<p>Total Staff</p>
<span><?php echo $total_staff; ?></span>
</div>
</div>
<canvas id="statsChart"></canvas>
</div>
<div class="dashboard-card">
<div class="dashboard-header">
<h1>Manage Users & Staff</h1>
</div>
<table class="user-table">
<thead>
<tr>
<th>Name</th>
<th>Email</th>
<th>Role</th>
<th>Actions</th>
<?php while ($user = $users->fetch_assoc()): ?>
<tr data-id="<?php echo $user['id']; ?>">
<td><input type="text" name="name" value="<?php echo
htmlspecialchars($user['name']); ?>" data-original="<?php echo
htmlspecialchars($user['name']); ?>" required></td>
<td><input type="email" name="email" value="<?php echo
htmlspecialchars($user['email']); ?>" data-original="<?php echo
htmlspecialchars($user['email']); ?>" required></td>
<td>
<select name="role" data-original="<?php echo $user['role']; ?>" required>
<option value="user" <?php echo $user['role'] == 'user' ? 'selected' : '';
?>>User</option>
<option value="staff" <?php echo $user['role'] == 'staff' ? 'selected' : '';
?>>Staff</option>
</select>

```

```

</td>
<td>
<button class="btn save-btn" onclick="saveUser(<?php echo $user['id'];
?>)">Save</button>
<button class="btn cancel-btn" onclick="cancelEdit(this)">Cancel</button>
<button class="btn delete-btn" onclick="showDeleteModal(<?php echo $user['id'];
?>)">Delete</button>
</td>
</tr>
<?php endwhile; ?>
</tbody>
</table>
<a href="#" class="add-btn" onclick="showAddModal()">Add New User/Staff</a>
</div>
<!-- Add User Modal -->
<div id="addModal" class="modal">
<div class="modal-content">
<h2>Add New User/Staff</h2>
<form method="POST">
<div class="form-group">
<label>Full Name</label>
<input type="text" name="name" required>
</div>
<div class="form-group">
<label>Email</label>
<input type="email" name="email" required>
</div>
<div class="form-group">
<label>Password</label>
<input type="password" name="password" required>
</div>
<div class="form-group">
<label>Role</label>
<select name="role" required>
<option value="user">User</option>
<option value="staff">Staff</option>
</select>
</div>
<button type="submit" name="add_user" class="btn save-btn">Add User</button>
<button type="button" onclick="hideAddModal()" class="btn cancel-
btn">Cancel</button>
</form>
</div>
</div>
<!-- Delete Confirmation Modal -->
<div id="deleteModal" class="modal">
<div class="modal-content">
<h2>Confirm Deletion</h2>
<p>Are you sure you want to delete this user?</p>
<form method="POST" id="deleteForm">
<input type="hidden" name="delete_id" id="deleteId">
<button type="submit" name="confirm_delete" class="btn delete-btn">Yes,
Delete</button>

```

```

</form>
</div>
</div>
</div>
<script>
// Chart.js for stats
const ctx = document.getElementById('statsChart').getContext('2d'); const statsChart =
new Chart(ctx, {
type: 'bar', data: {
labels: ['Total Users', 'Total Staff'], datasets: [{

label: 'User Statistics',
data: [<?php echo $total_users; ?>, <?php echo $total_staff; ?>], backgroundColor:
['#6366f1', '#f59e0b'],
borderColor: ['#4f46e5', '#d97706'], borderWidth: 1
}]
},
options: { scales: {
y: { beginAtZero: true }
},
plugins: {
legend: { display: false }
}
}
});
// Modal functions
function showAddModal() { document.getElementById('addModal').style.display = 'flex';
}
function hideAddModal() { document.getElementById('addModal').style.display = 'none';
}
function showDeleteModal(id) { document.getElementById('deleteId').value = id;
document.getElementById('deleteModal').style.display = 'flex';
}
function hideDeleteModal() { document.getElementById('deleteModal').style.display =
'none';
}
// Update user function function saveUser(id) {
const row = document.querySelector(`tr[data-id='${id}']`); const name =
row.querySelector('input[name="name"]').value; const email =
row.querySelector('input[name="email"]').value; const role =
row.querySelector('select[name="role"]').value;
const form = document.createElement('form'); form.method = 'POST';
form.style.display = 'none'; form.innerHTML = `
<input type="hidden" name="id" value="${id}">
<input type="hidden" name="name" value="${
// Cancel edit function function cancelEdit(button) { const row = button.closest('tr');
const fields = row.querySelectorAll('[name]'); fields.forEach(field => {
field.value = field.getAttribute('data-original');
});
}
</script>
</body>
</html>

```


CHAPTER – VI

TEST CASE

3.1 Introduction:

- The "Learning Spark" application is a web-based Learning Management System (LMS) designed to facilitate educational content delivery, real-time interaction, and administrative oversight efficiently.
- The system includes various modules such as user management, content management, blog management, chat system, and admin dashboard.
- This document outlines test cases to ensure the system functions as intended.

3.2 System Overview:

The "Learning Spark" application automates educational processes, ensuring seamless interaction between users (students), staff (educators), and administrators. It includes the following key functionalities:

- **User Management:** Handles student, staff, and admin accounts, including registration and authentication.
- **Content Management:** Enables staff to upload, update, and delete educational resources (videos/documents).
- **Blog Management:** Allows staff to post and manage daily blogs.
- **Chat System:** Facilitates real-time messaging between students, staff, and admin.
- **Admin Dashboard:** Provides tools for managing users, content, and system statistics.

3.3 TEST CASES

Test Case No. 1

TC1 - User Uploads Educational Content

Scenario: Verifies that staff can successfully upload educational content.

No	Procedure	Test Condition	Test Data	Expected Result
1	Access the content upload page.	Content upload page is accessible.	None	Content upload page is displayed.
2	Fill in content details.	Valid content details are entered.	Title: "Python Basics", Type: Video, File: python.mp4	Details are accepted.
3	Upload the content.	All required fields are filled.	Upload button clicked.	Content is stored successfully.
4	Verify content availability.	Content is saved in the system.	None	Content appears on user learning page.

Test Case No. 2

TC2 - Content Validation

Scenario: Checks if validation is applied correctly to uploaded content.

No	Procedure	Test Condition	Test Data	Expected Result
1	Submit new content.	New content details are available.	Title: "Java Intro", Type: Document, File: java.pdf	Validation is triggered.
2	Check validation status.	Content meets predefined criteria.	File size < 50MB, Format: .pdf/.mp4	Content is accepted or rejected based on criteria.

Test Case No. 3**TC3 - Student Notification for New Content**

Scenario: Ensures students receive notifications when new content is uploaded.

No	Procedure	Test Condition	Test Data	Expected Result
1	Upload new content to the database.	Content upload is completed.	Content ID: 4001, Title: "SQL Basics"	Notification is triggered.
2	Check notification delivery.	Notification is sent to users.	Message: "New content 'SQL Basics' uploaded."	Notification received by students.

TC4 - Admin Dashboard User Management

Scenario: Verifies that the admin can manage users effectively.

No	Procedure	Test Condition	Test Data	Expected Result
1	Log in to the admin dashboard.	Admin credentials are valid.	Email:jaga03038@gmail.com, Password: Tebi1328	Dashboard is displayed.
2	View user list.	Users exist in the system.	None	List of users (Admin/Staff/Students) is displayed.
3	Add a new user.	Valid user details are entered.	Name: "Jane Doe", Role: Staff, Email:pricy@gmail.com,12345678	New user is added successfully.

CHAPTER –VII

FUTURE ENHANCEMENT

1.FUTURE ENHANCEMENT

Future enhancements for the "Learning Spark" application include implementing AI-based content recommendations to provide personalized learning suggestions based on user interests, progress, and past interactions. Automated content tagging and smart profile building will allow the system to extract key topics from uploaded resources and suggest improvements to user profiles. A video lecture and live tutoring feature can be introduced, enabling staff to upload recorded lectures and schedule real-time tutoring sessions within the app. Advanced filtering and AI-powered search will enhance content discovery by offering insights into resource popularity and staff ratings. Integration with external educational platforms and open-source learning repositories will expand the range of available resources for users. Multi-language support will improve accessibility for a diverse student base. Blockchain-based verification can ensure secure authentication of certificates and user credentials earned through the platform. An AI-powered chatbot will offer instant assistance to students, staff, and admin, enhancing user engagement. A peer-to-peer collaboration and reward system can be introduced to incentivize knowledge sharing and active participation. Additionally, an admin dashboard with advanced analytics will provide insights into content usage, user performance, and system efficiency. These enhancements will make "Learning Spark" more efficient, interactive, and scalable for future growth.

CHAPTER – VIII

CONCLUSION

2. CONCLUSION

The "Learning Spark" application was designed with flexibility in mind, allowing for easy future modifications. The following conclusions can be deduced from the development of the project:

- This app can help students access educational resources and connect with educators effortlessly.
- We have considered the post-COVID educational landscape, which is why we developed this web-based app, accessible to anyone with an internet connection.
- Since most students and staff use mobile devices or laptops daily, it was practical to create a web app compatible with multiple browsers, rather than limiting it to a desktop-only platform, as many users no longer rely solely on traditional computers.
- Educational institutions, such as Don Bosco College, can use this app to provide structured learning opportunities to students remotely.
- Many students faced disruptions in education during the COVID situation, making "Learning Spark" a timely solution to support continuous learning.
- If an educational organization or institution approaches us, we can scale up the app and incorporate new features tailored to their needs, and we would be delighted to contribute to improving education accessibility.
- The system has sufficient scope for modification in the future if necessary.

CHAPTER – IX

REFERENCES

3.REFERENCES

Programming Languages & Development Platforms

- PHP: <https://www.php.net/> – Official PHP website for documentation, downloads, and updates.
- HTML/CSS/JavaScript: <https://developer.mozilla.org/en-US/> – Mozilla Developer Network for web development resources and standards.
- MySQL: <https://www.mysql.com/> – Official MySQL website for database documentation and downloads.

Backend Development & Tools

- Apache: <https://httpd.apache.org/> – Official Apache web server documentation and resources.